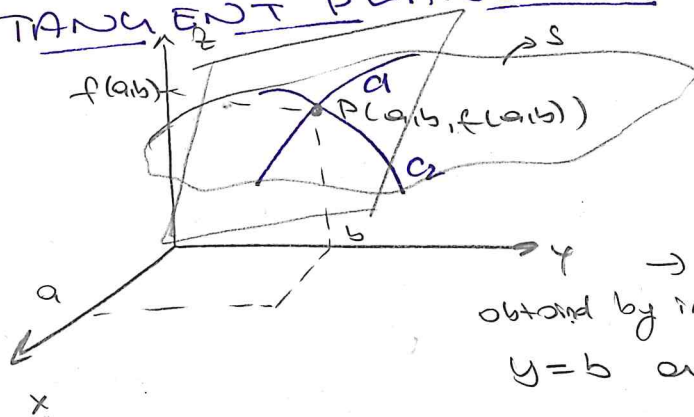


CALCULUS (II)

WEEK (IX)

TANGENT PLANES AND NORMAL LINES



Suppose a surface S has equation $z=f(x,y)$ where f has continuous partial derivatives and let $P(a,b,f(a,b))$ be a point on S .

Let C_1 and C_2 be the curves obtained by intersecting the vertical planes $y=b$ and $x=a$ with the surface S

Then, the point P lies on both C_1 and C_2 . Let T_1 and T_2 be the tangent lines to curves C_1 and C_2 at point P . Then, the tangent plane to the surface S at point P is defined to be the plane contains both lines T_1 and T_2

Tangent plane intersects the vertical plane $y=b$ in a straight line has the slope $f_1(a,b)$. [Remember any plane passing $P(a,b,f(a,b))$ has the form $A(x-a)+B(y-b)+C(z-f(a,b))=0$. $z-f(a,b)=-\frac{A}{C}(x-a)-\frac{B}{C}(y-b)$]
So we have $z-f(a,b)=f_1(a,b)(x-a)$

$$\vec{T}_1 = \vec{i} + f_1(a,b)\vec{k}$$

Similarly tangent plane intersects the vertical plane $x=a$ in a straight line has the slope $f_2(a,b)$ [Tangent line is $x=a$ and $z-f(a,b)=f_2(a,b)(y-b)$]

$$\vec{T}_2 = \vec{j} + f_2(a,b)\vec{k}$$

Then, the tangent plane to the surface S $z=f(x,y)$ at point P is defined to be the plane that contains both tangent lines T_1 and T_2

Normal of tangent plane perpendicular to both T_1

and T_2

$$\vec{n} = \vec{T}_2 \times \vec{T}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & f_2(a,b) \\ 1 & 0 & f_1(a,b) \end{vmatrix} = f_1(a,b)\vec{i} + f_2(a,b)\vec{j} - \vec{k}$$

Normal vector to $z=f(x,y)$ at $(a,b,f(a,b))$ is
$$\vec{n} = f_1(a,b)\vec{i} + f_2(a,b)\vec{j} - \vec{k}$$

An equation of the tangent plane to $z=f(x,y)$ at $(a,b,f(a,b))$ is

$$\boxed{z = f(a,b) + f_1(a,b)(x-a) + f_2(a,b)(y-b)}$$

Tangent plane

A normal vector to $z=f(x,y)$ at $(a,b,f(a,b))$ is

$$\boxed{\vec{n} = f_1(a,b)\vec{i} + f_2(a,b)\vec{j} - \vec{k}}$$

Normal vector

Normal line to $z=f(x,y)$ at $(a,b,f(a,b))$ has direction vector $f_1(a,b)\vec{i} + f_2(a,b)\vec{j} - \vec{k}$ and so has equation

$$\boxed{\frac{x-a}{f_1(a,b)} = \frac{y-b}{f_2(a,b)} = \frac{z-f(a,b)}{-1}}$$

Normal line

with suitable modification if either $f_1(a,b)=0$ or $f_2(a,b)=0$.

Ex 7 Find a normal vector and equations of the tangent plane and normal line to the graph $z = \sin(xy)$ at point where $x = \pi/3$ and $y = -1$

$$\frac{\partial z}{\partial x} = \cos(xy) \cdot y$$

$$\frac{\partial z}{\partial y} = \cos(xy) \cdot x$$

$$\left. \frac{\partial z}{\partial x} \right|_{(\pi/3, -1)} = -1/2$$

$$\left. \frac{\partial z}{\partial y} \right|_{(\pi/3, -1)} = \pi/6$$

$$\vec{n} = -1/2\vec{i} + \pi/6\vec{j} - \vec{k}$$

Tangent plane $z = \sin(xy) \Rightarrow z = \sin(-\pi/3) = -\frac{\sqrt{3}}{2}$

$$\boxed{z = -\frac{\sqrt{3}}{2} - \frac{1}{2}(x - \pi/3) + \pi/6(y + 1)}$$

Tangent plane

$$\boxed{\frac{x - \pi/3}{-1/2} = \frac{y + 1}{\pi/6} = \frac{z - (-\sqrt{3}/2)}{-1}}$$

Normal line

Ex 7 What horizontal plane is tangent to the surface.

$$z = x^2 - 4xy - 2y^2 + 12x - 12y - 1 \quad \text{and what is the point of}$$

tangency.

A plane is horizontal if its equation is of the form $z = k$, that is, it is independent of x and y . Therefore, we must have $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} = 0$ at the point of tangency. The equations,

$$\frac{\partial z}{\partial x} = 2x - 4y + 12 = 0$$

$$\frac{\partial z}{\partial y} = -4x - 4y - 12 = 0$$

} have solution $x = -4$
and $y = 1$. For these
values we have $z = -31$.

So, the required tangent plane has equation $z = -31$ and
the point of tangency is $(-4, 1, -31)$.

Ex 7 Find equation of the tangent plane to $z = \ln(2x+y)$ at $(-1, 3)$.

$$f(x, y) = \ln(2x+y) \quad f_x = \frac{2}{2x+y} \quad f_y = \frac{1}{2x+y}$$

$$f(-1, 3) = \ln(1) = 0 \quad f_x(-1, 3) = \frac{2}{1} = 2 \quad f_y(-1, 3) = \frac{1}{1} = 1$$

$$z = 0 + 2(x+1) + 1(y-3)$$

HIGHER-ORDER DERIVATIVES

$$z = f(x, y)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial z}{\partial x} = f_{11}(x, y) = f_{xx}(x, y)$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \frac{\partial z}{\partial y} = f_{22}(x, y) = f_{yy}(x, y)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \frac{\partial z}{\partial y} = f_{21}(x, y) = f_{yx}(x, y)$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \frac{\partial z}{\partial x} = f_{12}(x, y) = f_{xy}(x, y)$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \frac{\partial z}{\partial x} = f_{12}(x, y) = f_{xy}(x, y)$$

Similarly, if $w = f(x, y, z)$, then

$$\frac{\partial^5 w}{\partial y \partial x \partial y^2 \partial z} = \frac{\partial}{\partial y} \frac{\partial}{\partial x} \frac{\partial}{\partial y} \frac{\partial}{\partial y} \frac{\partial w}{\partial z} = f_{32212}(x, y, z) = f_{z4yx4}(x, y, z)$$

Ex 7 Find the four second partial derivatives of $f(x, y) = x^3 y^4$

$$\begin{aligned} f_1(x, y) &= 3x^2 y^4 & f_2(x, y) &= 4x^3 y^3 \\ f_{12}(x, y) &= 6x y^4 & f_{21}(x, y) &= 12x^2 y^3 \\ f_{12}(x, y) &= 12x^2 y^3 & f_{22}(x, y) &= 12x^3 y^2 \end{aligned}$$

Ex 7 Calculate $f_{223}(x, y, z)$, $f_{232}(x, y, z)$ and $f_{322}(x, y, z)$ for $f(x, y, z) = e^{x-2y+3z}$

$$\begin{aligned} f_{223} &= \frac{\partial}{\partial z} \frac{\partial}{\partial y} \frac{\partial}{\partial y} e^{x-2y+3z} = \frac{\partial}{\partial z} \frac{\partial}{\partial y} (-2e^{x-2y+3z}) \\ &= \frac{\partial}{\partial z} (4e^{x-2y+3z}) = 12e^{x-2y+3z} \end{aligned}$$

$$\begin{aligned} f_{322} &= \frac{\partial}{\partial y} \frac{\partial}{\partial y} \frac{\partial}{\partial z} e^{x-2y+3z} = \frac{\partial}{\partial y} \frac{\partial}{\partial y} (3e^{x-2y+3z}) \\ &= \frac{\partial}{\partial y} (-6e^{x-2y+3z}) = -12e^{x-2y+3z} \end{aligned}$$

$$f_{232} = \frac{\partial}{\partial y} \frac{\partial}{\partial z} \frac{\partial}{\partial y} e^{x-2y+3z} = \frac{\partial}{\partial y} \frac{\partial}{\partial z} (-2e^{x-2y+3z}) = \frac{\partial}{\partial y} (-6e^{x-2y+3z}) = -12e^{x-2y+3z}$$

Thm (Equality of mixed partials)

Suppose that two mixed n th-order partial derivatives of a function f involve the same differentiation but in different orders. If these partials are continuous at a point P and if all partials of f of order less than n are continuous in a neighborhood of P , then two mixed partials are equal at point P .

Euler Theorem

(Clairaut's Theorem)

Suppose f is defined on a disk D that contains the point (a,b) . If the functions f_x, f_y, f_{xy} and f_{yx} are all continuous on D , then

$$f_{xy}(a,b) = f_{yx}(a,b)$$

Ex Find the second derivatives of the following functions

① $f(x,y) = \sin(xy)$

$$f_x(x,y) = \cos(xy) \cdot y$$

$$f_y(x,y) = \cos(xy) \cdot x$$

$$f_{xx}(x,y) = -\sin(xy) \cdot y^2$$

$$f_{xy}(x,y) = -\sin(xy) \cdot xy + \cos(xy)$$

$$f_{yy}(x,y) = -\sin(xy) \cdot x^2$$

$$f_{yx}(x,y) = -\sin(xy) \cdot xy + \cos(xy)$$

Note $f_{xy}(x,y) = f_{yx}(x,y)$

② $g(x,y) = x^2y + \cos y + y \sin x$

$$g_x = 2xy + y \cos x$$

$$g_{xx} = 2y - y \sin x$$

$$g_{xy} = 2x + \cos x$$

$$g_y = x^2 - \sin y + \sin x$$

$$g_{yx} = 2x + \cos x$$

$$g_{yy} = -\cos y$$

③ $f(x,y) = x^3 + x^2y^3 - 2y^2$

$$f_x = 3x^2 + 2xy^3$$

$$f_{xx} = 6x + 2y^3$$

$$f_{xy} = 6xy^2$$

$$f_y = 3x^2y^2 - 4y$$

$$f_{yy} = 6x^2y - 4$$

$$f_{yx} = 6xy^2$$

THE CHAIN RULE

(I) $z = f(u(t), v(t)) = f(t)$

$$g'(t) = \lim_{h \rightarrow 0} \frac{g(t+h) - g(t)}{h} = \lim_{h \rightarrow 0} \frac{f(u(t+h), v(t+h)) - f(u(t), v(t))}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(u(t+h), v(t+h)) - f(u(t), v(t+h))}{h} + \lim_{h \rightarrow 0} \frac{f(u(t), v(t+h)) - f(u(t), v(t))}{h}$$

$$= f_1(u(t), v(t)) u'(t) + f_2(u(t), v(t)) v'(t)$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\lim_{h \rightarrow 0} \frac{f(u(t), v(t+h)) - f(u(t), v(t))}{h} = \lim_{h \rightarrow 0} \frac{v(t+h) - v(t)}{h} \cdot \lim_{h \rightarrow 0} \frac{f(u(t), v(t+h)) - f(u(t), v(t))}{v(t+h) - v(t)}$$

$$\lim_{h \rightarrow 0} \frac{f(u(t+h), v(t+h)) - f(u(t), v(t+h))}{u(t+h) - u(t)} \cdot \lim_{h \rightarrow 0} \frac{u(t+h) - u(t)}{h} + \lim_{h \rightarrow 0} \frac{f(u(t), v(t+h)) - f(u(t), v(t))}{v(t+h) - v(t)} \cdot \lim_{h \rightarrow 0} \frac{v(t+h) - v(t)}{h}$$

(II) $z = f(x, y) \quad x = q(s, t) \quad y = b(s, t)$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Ex 7 If $z = x^2y + 3xy^4$ where $x = \sin 2t$ and $y = \cos t$ find $\frac{dz}{dt}$ for $t = \frac{\pi}{4}$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = (2xy + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t)$$

$$\frac{dz}{dt} = (2 \sin 2t \cos t + 3 \cos^4 t)(2 \cos 2t) - (\sin^2 2t + 12 \sin 2t \cos^3 t) \sin t$$

$$\left. \frac{dz}{dt} \right|_{t=\frac{\pi}{4}} = 6$$

→ This derivative can be interpreted as the rate of change of z w.r.t t as the point (x, y) moves along the curve C with parametric equations $x = \sin 2t, y = \cos t$.

Ex 7 If $g(s,t) = f(s^2 - t^2, t^2 - s^2)$ and f is differentiable show that g satisfies $t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0$

let $x = s^2 - t^2$, $y = t^2 - s^2$. Then, $g(s,t) = f(x,y)$

$$\frac{\partial g}{\partial s} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial s} = 2s \frac{\partial f}{\partial x} - 2s \frac{\partial f}{\partial y} \quad / \text{multiply by } t$$

$$\frac{\partial g}{\partial t} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial t} = -2t \frac{\partial f}{\partial x} - 2t \frac{\partial f}{\partial y} \quad / \text{multiply by } s$$

$$t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0$$

Ex 7 If $z = f(x,y)$ has continuous second order partial derivatives and $x = r^2 + s^2$ and $y = rs$ find $\frac{\partial^2 z}{\partial r^2}$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = 2r z_x + s z_y$$

$$\frac{\partial^2 z}{\partial r^2} = \frac{\partial}{\partial r} (2r z_x + s z_y) = 2 z_x + 2r \frac{\partial^2 z}{\partial r \partial x} + s \frac{\partial^2 z}{\partial r \partial y}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial r \partial x} &= \frac{\partial}{\partial r} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) \frac{\partial x}{\partial r} + \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) \frac{\partial y}{\partial r} \\ &= z_{xx} \frac{2r}{\cancel{\frac{\partial x}{\partial r}}} + z_{xy} \frac{s}{\cancel{\frac{\partial y}{\partial r}}} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial r \partial y} &= \frac{\partial}{\partial r} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) \frac{\partial x}{\partial r} + \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) \frac{\partial y}{\partial r} \\ &= z_{xy} (2r) + z_{yy} (s) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial r^2} &= 2 z_x + 2r (z_{xx} 2r + s z_{xy}) + s (z_{xy} 2r + z_{yy} s) \\ &= 2 z_x + 4r^2 z_{xx} + 4rs z_{xy} + s^2 z_{yy} \end{aligned}$$

Ex ① $z = x^2y + xy^2$ $x = 2+t^4$ $y = 1-t^3$ $\frac{dz}{dt} = ?$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = (2xy + y^2)(3t^3) + (x^2 + 2xy)(-3t^2)$$

$$= (2(2+t^4)(1-t^3) + (1-t^3)^2)3t^3 + ((2+t^4)^2 + 2(2+t^4)(1-t^3))(-3t^2)$$

② $z = \arctan(2x+y)$ $x = s^2t$ $y = slt$ $\frac{\partial z}{\partial s} = ?$ $\frac{\partial z}{\partial t} = ?$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \frac{2}{1+(2x+y)^2} \cdot 2st + \frac{1}{1+(2x+y)^2} \cdot lt$$

$$= \frac{2st + lt}{1+(2s^2t + slt)^2}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= \frac{2}{1+(2x+y)^2} \cdot s^2 + \frac{1}{1+(2x+y)^2} \cdot \frac{s}{t}$$

$$= \frac{2s^2}{1+(2s^2t + slt)^2} + \frac{s}{1+(2s^2t + slt)^2} \cdot \frac{1}{t}$$

③ $\frac{\partial}{\partial y} f(y, g(x,t), g(y,t)) = ?$ $u = g(x,t)$
 $v = g(y,t)$

$$\frac{\partial}{\partial y} (f(u,v)) = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y}$$

$$= f_u g(x,t) + f_v \frac{\partial g}{\partial y}$$

Ex 7. $z = \sin(x^2y)$ where $x = st^2$ $y = s^2 + 1/t$ find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t} = ?$

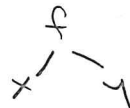
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= (2xy \cos(x^2y)) (2st) + (x^2 \cos(x^2y)) \left(-\frac{1}{t^2}\right)$$

$$= (2st^2 (s^2 + \frac{1}{t}) 2st + s^2 t^4 (-\frac{1}{t^2})) \cos(s^4 t^4 + s^2 t^3)$$

$$= (4s^4 t^3 + 3s^2 t^2) \cos(s^4 t^4 + s^2 t^3)$$

Ex 7 Find $\frac{\partial}{\partial x} f(x^2y, x+2y)$ and $\frac{\partial}{\partial y} f(x^2y, x+2y)$ in terms of partial derivatives of f



$$\begin{aligned} \frac{\partial}{\partial x} f(x^2y, x+2y) &= f_1(x^2y, x+2y) \frac{\partial}{\partial x} (x^2y) + f_2(x^2y, x+2y) \frac{\partial}{\partial y} (x+2y) \\ &= 2xy f_1(x^2y, x+2y) + f_2(x^2y, x+2y) \end{aligned}$$

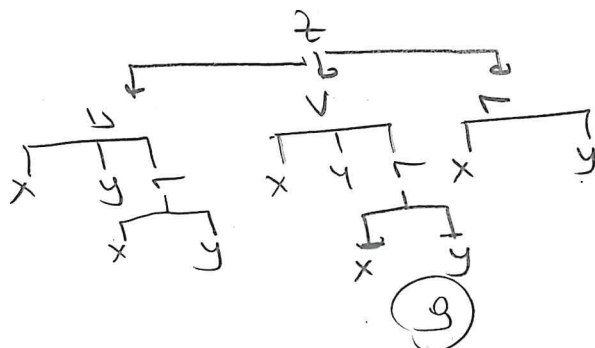
$$\begin{aligned} \frac{\partial}{\partial y} f(x^2y, x+2y) &= f_1(x^2y, x+2y) \frac{\partial}{\partial y} (x^2y) + f_2(x^2y, x+2y) \frac{\partial}{\partial y} (x+2y) \\ &= x^2 f_1(x^2y, x+2y) + 2 f_2(x^2y, x+2y) \end{aligned}$$

Ex 7 Express the partial derivatives of $z = h(s, t) = f(g(s, t))$ in terms of derivative f' of f and partial derivative of g

$$h_1(s, t) = \frac{\partial z}{\partial s} = \frac{dz}{dx} \frac{\partial x}{\partial s} = f'(g(s, t)) g_1(s, t)$$

$$h_2(s, t) = \frac{\partial z}{\partial t} = \frac{dz}{dx} \frac{\partial x}{\partial t} = f'(g(s, t)) g_2(s, t)$$

Ex 7 Write the appropriate version of Chain Rule for $\frac{\partial z}{\partial x}$ where z depends on u, v and r ; u and v depend on x, y and r ; and r depends on x and y



$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial z}{\partial r} \frac{\partial r}{\partial x} \\ &+ \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial z}{\partial r} \frac{\partial r}{\partial y} \\ &+ \frac{\partial z}{\partial r} \frac{\partial r}{\partial x} \end{aligned}$$

Homogeneous Functions

A function $f(x_1, x_2, \dots, x_n)$ is said to be positively homogeneous of degree k if, for every point $(x_1, x_2, \dots, x_n)$ in its domain and every real number $t > 0$, we have

$$f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n)$$

Ex 7

$f(x, y) = x^2 + xy - y^2$	positively homogeneous of degree 2
$f(x, y) = \sqrt{x^2 + y^2}$	positively homogeneous of degree 1.
$f(x, y) = \frac{2xy}{x^2 + y^2}$	" " " " " "

$f(x, y, z) = \frac{x - y + 5z}{y^2 - z^2}$ positively homogeneous of degree -1

$f(x, y) = x^2 + y$ NOT positively homogeneous

Thm Euler Theorem

If $f(x_1, x_2, \dots, x_n)$ has continuous first partial derivatives and is positively homogeneous of degree k , then

$$\sum_{i=1}^n x_i f_i(x_1, x_2, \dots, x_n) = k f(x_1, \dots, x_n)$$

Proof Differentiate $f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n)$

w.r.t. t we get

$$x_1 f_1(tx_1, tx_2, \dots, tx_n) + x_2 f_2(tx_1, \dots, tx_n) + \dots + x_n f_n(tx_1, \dots, tx_n) = k t^{k-1} f(x_1, x_2, \dots, x_n)$$

Ex 7 Calculate $\frac{\partial^2}{\partial x \partial y} f(x^2 - y^2, xy)$ in terms of partial derivatives of f . Assume second order partials of f are cont.

$\frac{\partial^2}{\partial x \partial y} f(u, v) \quad u = x^2 - y^2 \quad v = xy$

$$\frac{\partial}{\partial y} f(u, v) = -2y f_1(u, v) + x f_2(u, v)$$

$$\frac{\partial^2}{\partial x \partial y} f(u, v) = -2y (2x f_{11}(u, v) + y f_{12}(u, v))$$

$$+ f_2(u, v) + x (2x f_{21}(u, v) + y f_{22}(u, v))$$

$$= f_2(u, v) - 4xy f_{11}(u, v) + 2(x^2 - y^2) f_{12}(u, v) + xy f_{22}(u, v)$$

IMPLICIT FUNCTIONS & IMPLICIT DIFFERENTIATION

Implicit Functions

→ Let F be a differentiable function of two variables x and y , that is, $F(x, y) = 0$.

→ Suppose that any point (a, b) satisfies the equation. $F(a, b) = 0$ and suppose $F(x, y) = 0$ can be solved for y as a function of x . Differentiating w.r.t. x gives

$$F_1(x, y) + F_2(x, y) \frac{dy}{dx} = 0$$

Thus, $\left. \frac{dy}{dx} \right|_{x=a} = - \frac{F_1(a, b)}{F_2(a, b)}$ if $F_2(a, b) \neq 0$.

→ observe that the condition $F_2(a, b) \neq 0$ required for the calculation of $y'(a)$ will itself guarantee that the solution $y(x)$ exist.

→ $F_2(a, b) \neq 0$ and differentiability of $F(x, y)$ near (a, b) implies the level curve $F(x, y) = F(a, b)$ has a nonvertical tangent line near (a, b) .

→ Similarly, let $F(x, y, z) = 0$ and $z = z(x, y)$

$$F_1(x, y, z) \frac{dx}{dx} + F_2(x, y, z) \frac{dy}{dx} + F_3(x, y, z) \frac{\partial z}{\partial x} = 0$$

$$F_1(x, y, z) \frac{dx}{dy} + F_2(x, y, z) \frac{dy}{dy} + F_3(x, y, z) \frac{\partial z}{\partial y} = 0$$

Then,

$$F_1(x, y, z) + F_3(x, y, z) \frac{\partial z}{\partial x} = 0$$

$$F_2(x, y, z) + F_3(x, y, z) \frac{\partial z}{\partial y} = 0$$

(11)

$$\frac{\partial z}{\partial x} = - \frac{F_1(x, y, z)}{F_3(x, y, z)}$$
$$\frac{\partial z}{\partial y} = - \frac{F_2(x, y, z)}{F_3(x, y, z)}$$

$$\left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)} = - \frac{F_1(x_0, y_0, z_0)}{F_3(x_0, y_0, z_0)}, \quad \left. \frac{\partial z}{\partial y} \right|_{(x_0, y_0)} = - \frac{F_2(x_0, y_0, z_0)}{F_3(x_0, y_0, z_0)}$$

If $F_3(x_0, y_0, z_0) \neq 0$.

Ex Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the following functions.

(a) $x^3 + y^3 + z^3 + bxyz = 1$ $F(x, y, z) = x^3 + y^3 + z^3 + bxyz - 1$

$$\frac{\partial z}{\partial x} = - \frac{F_x}{F_z} = - \frac{3x^2 + byz}{3z^2 + bxy}$$

$$\frac{\partial z}{\partial y} = - \frac{F_y}{F_z} = - \frac{(3y^2 + bxz)}{(3z^2 + bxy)}$$

(b) $yz = \ln(x+z)$ $F(x, y, z) = yz - \ln(x+z)$

$$\frac{\partial z}{\partial x} = - \frac{F_x}{F_z} = \frac{-1/(x+z)}{y - 1/(x+z)} = - \frac{1}{y(x+z) - 1}$$

$$\frac{\partial z}{\partial y} = - \frac{F_y}{F_z} = \frac{z}{y - 1/(x+z)} = \frac{z(x+z)}{(x+z)y - 1}$$

Ex Near what points on the sphere $x^2 + y^2 + z^2 = 1$ can the equation of the sphere be solved for z as a function of x and y ? Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at such points.

The sphere is the level surface $F(x, y, z) = 0$, where

$$F(x, y, z) = x^2 + y^2 + z^2 - 1$$

$$F_3(x, y, z) = \frac{\partial F}{\partial z} = 2z = 0 \Rightarrow z = 0$$

Thus the eq. $F(x, y, z) = 0$ can be solved for $z = z(x, y)$ near P_0 if P_0 doesn't lie on the circle

$$x^2 + y^2 = 1, z = 0.$$

→ If $z \neq 0$, we can calculate the partial derivatives of the solution $z = z(x, y)$ by implicit differentiating the equation of the sphere $x^2 + y^2 + z^2 = 1$

$$2x + 2z \frac{\partial z}{\partial x} = 0 \quad \text{so,} \quad \frac{\partial z}{\partial x} = -\frac{x}{z}$$

$$2y + 2z \frac{\partial z}{\partial y} = 0 \quad \text{so} \quad \frac{\partial z}{\partial y} = -\frac{y}{z}$$

→ When we are given $F(x, y, z) = 0$ and asked $\frac{\partial x}{\partial z}$, we could understand that x is intended to be a function of y and z that is $x = x(y, z)$

→ What if we are given a system

$$F(x, y, z, w) = 0$$

$$G(x, y, z, w) = 0$$

and asked $\frac{\partial x}{\partial z} = ?$ In short, which of the situations

$$\left\{ \begin{array}{l} x = x(z, w) \\ y = y(z, w) \end{array} \right. \quad \text{and} \quad \left\{ \begin{array}{l} x = x(y, z) \\ w = w(y, z) \end{array} \right\}$$

are we dealing with?

We use subscripts to specify which variable is to be regarded as the other independent variable.

$$\left(\frac{\partial x}{\partial z} \right)_w \quad \text{implies the interpretation} = \left\{ \begin{array}{l} x = x(z, w) \\ y = y(z, w) \end{array} \right.$$

$$\left(\frac{\partial x}{\partial z} \right)_y \quad \text{implies the interpretation} = \left\{ \begin{array}{l} x = x(y, z) \\ w = w(y, z) \end{array} \right.$$

Ex Given equations $F(x,y,z,w) = 0$ and $G(x,y,z,w) = 0$

calculate $\left(\frac{\partial x}{\partial z}\right)_w$

$$x = x(z, w) \quad y = y(z, w)$$

$$F_1 \frac{\partial x}{\partial z} + F_2 \frac{\partial y}{\partial z} + F_3 = 0$$

$$G_1 \frac{\partial x}{\partial z} + G_2 \frac{\partial y}{\partial z} + G_3 = 0$$

Eliminate $\frac{\partial y}{\partial z}$, we obtain

$$\left(\frac{\partial x}{\partial z}\right)_w = -\frac{F_3 G_2 - F_2 G_3}{F_1 G_2 - F_2 G_1}$$

Ex let x, y, u and v be related equations

$$u = x^2 + xy - y^2 \quad \textcircled{a} \left(\frac{\partial x}{\partial u}\right)_y \quad \text{at } (x, y) = (2, -1)$$

$$v = 2xy + y^2 \quad \textcircled{b} \left(\frac{\partial x}{\partial v}\right)_y \quad \text{at } (x, y) = (2, -1)$$

$$\textcircled{a} \quad x = x(u, v) \quad y = y(u, v)$$

$$1 = \frac{\partial u}{\partial u} = 2x \cdot \frac{\partial x}{\partial u} + y \cdot \frac{\partial x}{\partial u} + x \frac{\partial y}{\partial u} - 2y \cdot \frac{\partial y}{\partial u}$$

$$1 = (2x + y) \frac{\partial x}{\partial u} + (x - 2y) \frac{\partial y}{\partial u}$$

$$0 = \frac{\partial v}{\partial u} = 2y \cdot \frac{\partial x}{\partial u} + 2x \frac{\partial y}{\partial u} + 2y \cdot \frac{\partial y}{\partial u}$$

$$= 2y \frac{\partial x}{\partial u} + (2x + 2y) \frac{\partial y}{\partial u}$$

$$\text{At } (x, y) = (2, -1)$$

$$1 = 3 \frac{\partial x}{\partial u} + 4 \frac{\partial y}{\partial u}$$

$$0 = -2 \frac{\partial x}{\partial u} + 2 \frac{\partial y}{\partial u}$$

Eliminate $\frac{\partial y}{\partial u}$ gives

$$\left(\frac{\partial x}{\partial u}\right)_y = 1/7$$

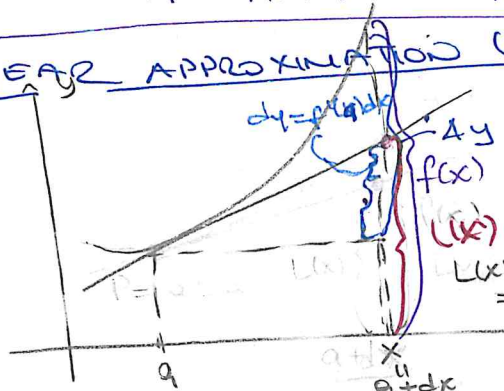
⑤ $x = x(u, y)$
 $v = v(u, y)$

$$\left. \begin{aligned} \frac{\partial u}{\partial v} &= 1 = (2x+y) \frac{\partial x}{\partial u} \\ \frac{\partial v}{\partial u} &= 2y \cdot \frac{\partial x}{\partial u} \end{aligned} \right\} (x, y) = (2, -1)$$

$$\left(\frac{\partial x}{\partial u} \right)_y = \frac{1}{3}$$

[JACOBIAN DETERMINANTS, IMPLICIT FUNCTION THEOREM IN ADAMS (NOT GIVEN)]
LINEAR APPROXIMATIONS, DIFFERENTIABILITY AND DIFFERENTIAL

LINEAR APPROXIMATION (TANGENT PLANE APPROXIMATION)



$$f(x) \approx L(x) = f(a) + f'(a)(x-a)$$

Here, $L(x)$ is the linearization

of f at a , its graph is the tangent line to $y=f(x)$, there

→ Similarly, the tangent plane to the graph of $z=f(x, y)$ and (a, b) is $z=L(x, y)$ where

$$L(x, y) = f(a, b) + f_1(a, b)(x-a) + f_2(a, b)(y-b) \rightarrow \text{Tangent plane at } (a, b)$$

is the linearization of f at (a, b) . We can use $L(x, y)$ to approximate values of $f(x, y)$

$$f(x, y) \approx L(x, y) = f(a, b) + f_1(a, b)(x-a) + f_2(a, b)(y-b)$$

ex Find an approximate value for $f(x, y) = \sqrt{2x^2 + e^{2y}}$ at $(2.2, -0.2)$.

Let $(a, b) = (2, 0)$ $f(2, 0) = 3$

$$f_1(x, y) = \frac{\partial}{\partial x} \sqrt{2x^2 + e^{2y}} \quad f_1(2, 0) = 3$$

$$f_2(x, y) = \frac{\partial}{\partial y} \sqrt{2x^2 + e^{2y}} \quad f_2(2, 0) = \frac{4}{3}$$

Thus, $L(x, y) = 3 + \frac{4}{3}(x-2) + \frac{1}{3}(y-0)$ or

$$f(2.2, -0.2) \approx L(2.2, -0.2) = 3 + \frac{4}{3}(2.2-2) + \frac{1}{3}(-0.2-0) = 3.1$$

$f(2.2, -0.2) \approx 3.17$ (15 decimal places)

Ex 7 Find the linear approximation to $A = 3 + \frac{x^2}{16} + \frac{y^2}{9}$ at $(-4, 3)$

$$f(x, y) = 3 + \frac{x^2}{16} + \frac{y^2}{9} \quad f_x(x, y) = \frac{x}{8} \Rightarrow f_x(-4, 3) = -\frac{1}{2}$$

$$f_y(x, y) = \frac{2y}{9} \Rightarrow f_y(-4, 3) = \frac{2}{3}$$

The tangent plane or linear approximation is

$$L(x, y) = 5 - \frac{1}{2}(x+4) + \frac{2}{3}(y-3)$$

Ex 7 Find the linearization of

$$f(x, y) = x^2 - xy + \frac{1}{2}y^2 + 3 \text{ at point } (3, 2)$$

$$f(3, 2) = 9 - 6 + \frac{1}{2} \cdot 4 + 3 = 8$$

$$\frac{\partial f}{\partial x} = -x + y$$

$$\frac{\partial f}{\partial y} \Big|_{(3, 2)} = -3 + 2 = -1$$

$$\frac{\partial f}{\partial x} = 2x - y$$

$$\frac{\partial f}{\partial x} \Big|_{(3, 2)} = 4$$

$$L(x, y) = f(3, 2) + f_x(3, 2)(x-3) + f_y(3, 2)(y-2)$$

Ex 7 Find approximate value for $f(2.1, 2.9) = \sqrt{41 - 4(2.1)^2 - (2.9)^2}$ to approximate $f(2.1, 2.9)$ we can use point $(a, b) = (2, 3)$

$$f(x, y) = \sqrt{41 - 4x^2 - y^2}$$

$$f(2, 3) = \sqrt{41 - 4 \cdot 4 - 9} = 4$$

$$f_x(x, y) = \frac{-4x}{\sqrt{41 - 4x^2 - y^2}}$$

$$f_x(2, 3) = -2$$

$$f_y(x, y) = \frac{-y}{\sqrt{41 - 4x^2 - y^2}}$$

$$f_y(2, 3) = -\frac{3}{4}$$

$$\begin{aligned} \text{Error} &= |f(x, y) - L(x, y)| \\ &= |f(2.1, 2.9) - L(2.1, 2.9)| \\ &= 0.019 \approx 0.02 \end{aligned}$$

$$L(x, y) = f(2, 3) + f_x(2, 3)(x-2) + f_y(2, 3)(y-3)$$

$$= 4 - 2(x-2) - \frac{3}{4}(y-3)$$

$$= \frac{41}{4} - 2x - \frac{3}{4}y$$

$$f(2.1, 2.9) \approx L(2.1, 2.9)$$

$$L(2.1, 2.9) = \frac{41}{4} - 2(2.1) - \frac{3}{4}(2.9)$$

$$R = f(2.1, 2.9) - \sqrt{41 - 4(2.1)^2 - (2.9)^2} \approx 3.8665 - 3.875 = -0.0085$$

DIFFERENTIABILITY

→ Unlike the single variable case, the mere existence of the partial derivatives $f_1(a,b)$ and $f_2(a,b)$ doesn't even imply that f is continuous at (a,b) . Moreover, it does not ensure that the error in linearization is small compared to distance $\sqrt{(x-a)^2 + (y-b)^2}$ between (a,b) and (x,y) .

Definition We say that the function $f(x,y)$ is differentiable

at the point (a,b) if

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(a+h, b+k) - f(a,b) - hf_1(a,b) - kf_2(a,b)}{\sqrt{h^2 + k^2}} = 0$$

In other words $\lim_{(x,y) \rightarrow (a,b)} \frac{|f(x,y) - L(x,y)|}{\|(x,y) - (a,b)\|} = 0$ where $a+h=x$
 $b+k=y$
 $\|(x,y) - (a,b)\| = \sqrt{(x-a)^2 + (y-b)^2} = \sqrt{h^2 + k^2}$

The function $f(x,y)$ is differentiable at the point (a,b)

if and only if the surface $z=f(x,y)$ has a nonvertical tangent plane at (a,b) . This implies that $f_1(a,b)$ and $f_2(a,b)$ must exist and that f must be continuous

Recall however, the existence of partial derivatives doesn't even imply that f is continuous, let alone differentiable

OR WE CAN SAY

Having partial derivatives doesn't even mean f is continuous, so of course we can't expect it to be differentiable either

let alone something stronger & more demanding than what came before, to emphasize how unlikely it is under just continuity. (let alone = bring you down, after mentioning it, hole up high.)

Thm If f_1 and f_2 are continuous in a neighbourhood of the point (a,b) , then f is differentiable at (a,b)

(★) Differentiable function is a function such that the linear approximation is good when (x,y) is near to (a,b) .

Ex Calculate $f(x+h, y+k) - f(x, y) - f_1(x, y)h - f_2(x, y)k$ if $f(x, y) = x^3 + xy^2$

Since $f_1(x, y) = 3x^2 + y^2$ and $f_2(x, y) = 2xy$, we have.

$$f(x+h, y+k) - f(x, y) - f_1(x, y)h - f_2(x, y)k = (x+h)^3 + (x+h)(y+k)^2 - x^3 - xy^2 - (3x^2 + y^2)h - 2xyk = 3xh^2 + h^3 + 2yhk + hk^2 + xk^2$$

$$\lim_{(h, k) \rightarrow (0, 0)} \frac{3xh^2 + h^3 + 2yhk + hk^2 + xk^2}{\sqrt{h^2 + k^2}} = 0$$

Ex Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its linearization there. Then, use it to approximate $f(1.1, -0.1)$

$$f_x = e^{xy} + xye^{xy} \quad f_y = x^2 e^{xy} \rightarrow \text{are continuous so } f \text{ is differentiable.}$$

$$f_x(1, 0) = 1 \quad f_y(1, 0) = 1$$

$$L(x, y) = f(1, 0) + f_x(1, 0)(x-1) + f_y(1, 0)(y-0) = x + y$$

$$xe^{xy} \approx x + y \quad f(1.1, -0.1) \approx 1.1 - 0.1 = 1$$

Indeed the actual value of $f(1.1, -0.1) = 1.1e^{-0.11} \approx 0.98542$

THM Differentiability Implies Continuity

If a function $f(x, y)$ is differentiable at (a, b) then f is continuous at (a, b) .
Differentiable $\Rightarrow$ Continuous.

PROOF Define $R(x, y) = f(x, y) - f(a, b) - f_1(a, b)(x-a) - f_2(a, b)(y-b)$

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) - f(a, b) = \lim_{(x, y) \rightarrow (a, b)} R(x, y) + f_1(a, b)(x-a) + f_2(a, b)(y-b)$$

Multiply and divide by $\sqrt{(x-a)^2 + (y-b)^2}$

$$\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - f(a, b)}{\sqrt{(x-a)^2 + (y-b)^2}} = \lim_{(x, y) \rightarrow (a, b)} \frac{R(x, y)}{\sqrt{(x-a)^2 + (y-b)^2}} + \lim_{(x, y) \rightarrow (a, b)} \frac{f_1(a, b)(x-a) + f_2(a, b)(y-b)}{\sqrt{(x-a)^2 + (y-b)^2}}$$

Mean that $\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - f(a, b)}{\sqrt{(x-a)^2 + (y-b)^2}} = 0$

DIFFERENTIALS

→ If the first partial derivatives of a function $z = f(x_1, x_2, \dots, x_n)$ exist at a point, we may construct a differential dz or df of the function at that point in a manner similar to.

that used functions of one variable.

$$dz = df = \frac{\partial z}{\partial x_1} dx_1 + \frac{\partial z}{\partial x_2} dx_2 + \dots + \frac{\partial z}{\partial x_n} dx_n$$

$$= f_{x_1}(x_1, x_2, \dots, x_n) dx_1 + \dots + f_{x_n}(x_1, x_2, \dots, x_n) dx_n.$$

→ For two dim. $z = f(x, y)$ $\boxed{dz = f_x(a, b)(x-a) + f_y(a, b)(y-b)}$
 $\boxed{f(x, y) \approx L(x, y) = f(a, b) + dz}$

→ For a differentiable function f , the differential df is an approximation the the change Δf in the value of the function given by

$$\Delta f = f(x_1 + dx_1, x_2 + dx_2, \dots, x_n + dx_n) - f(x_1, x_2, \dots, x_n)$$

$$\frac{\Delta f - df}{\sqrt{(dx_1)^2 + \dots + (dx_n)^2}} \rightarrow 0 \text{ as } dx_i \rightarrow 0 \quad (1 \leq i \leq n)$$

For two dim $z = f(x, y)$

$$\boxed{\Delta z = f(x+dx, y+dy) - f(x, y)}$$

$$\boxed{dz = f_x dx + f_y dy}$$

Ex Estimate the percentage change in the period $T = 2\pi \sqrt{\frac{L}{g}}$ of a simple pendulum if the length, L , of the pendulum increases by 2% ($\pm 2/100$) and the acceleration of gravity, g , decreases by 0.6% ($-6/1000$).

Calculate $dT = \frac{\partial T}{\partial L} dL + \frac{\partial T}{\partial g} dg$

$$= \frac{2\pi}{2\sqrt{Lg}} dL - \frac{2\pi\sqrt{L}}{2g^{3/2}} dg$$

$$dL = \frac{2}{100} L \quad dg = -\frac{6}{1000} g$$

$$dT = \frac{1}{100} 2\pi \sqrt{\frac{L}{g}} - \frac{2\pi\sqrt{L}}{2g^{3/2}} \left(-\frac{6}{1000} g\right)$$

$$= \frac{2\pi}{100} \sqrt{\frac{L}{g}} + \frac{6}{1000} \pi \sqrt{\frac{L}{g}} = \frac{13}{1000} T.$$

Ex 7 for $z = f(x, y) = x^2 + 3xy - y^2$ if x changes from 2 to 2.05 and y changes from 3 to 2.96 compare Δz and dz

$$f_x = 2x + 3y \quad f_y = 3x - 2y$$

$$dx = 2.05 - 2 = 0.05$$

$$dy = 2.96 - 3 = -0.04$$

$$dz = f_x(2, 3)dx + f_y(2, 3)dy = 13 \cdot 0.05 = 0.65$$

$$\Delta z = f(2.05, 2.96) - f(2, 3) = 0.6449$$

Note that $\Delta z \approx dz$.